



COMSATS University, Islamabad

(CUI)

Department of Computer Science

Quiz-2 Database Systems

Time Allowed: 20 minutes

Total Marks: 10

Class :BCS-3 Spring-2026

Date: 16 March 2026

NAME: _____ - _____ Reg.no: _____
Mapped to CLO-2 ((Using the concepts of relational **Algebra**.)

Q-1. Consider the following relation schema and Write the following queries in relational Algebra. [4]

Depositor(customername,accountnumber)

Borrower(customername,loanno)

Loan(loanno,branchname,amount)

- a. Find the names of the customers who have loan at Chennai branch.
- b. Find customers who have taken loans from all branches.
- c. Find loan numbers associated with customers who do not have an account.
- d. Find customers who have taken a loan from the same branch where they have an account.

Q2. Suppose relations R(A,B) and S(B,C,D) have the following tuples: [1]

A	B
1	2
3	4
5	6

B	C	D
2	4	6
4	6	8
4	7	9

Compute $R \bowtie_{\theta} S$ with the condition $R.A < S.C$ AND $R.B < S.D$.

Q3. Write a relational algebra expression using cross product to obtain the same result (Same number of columns and rows) as in Q2. [1]

Q4. Consider the following relations.[3]Product

Products_Detail Table

Prod_ID	Prod_Name
1	PRO11
2	PRO22
3	PRO34
4	PRO51
5	PRO64
6	PRO74

Prod_ID	Price	Qty	Defect_Rate
1	4000	25	12
2	3000	40	2
3	4000	35	8
4	3000	45	14
5	4000	30	0

Sold Table

Quality Table

Quality_Index	Min_Defect_Rate	Max_Defect_R
1	0	4
2	5	10
3	11	100

Prod_ID	Date	Qty _{sold}
1	12-Aug	45
2	12-Aug	5
3	12-Aug	300
1	13-Aug	25
2	13-Aug	4
3	13-Aug	75

Construct relational algebra queries for the following statements below:

1. Display the Product ID, Product Name and Price and Quantity from the product and product details tables.
2. Display the Product ID, Product Name and Quantity of only those products whose price is 4000 rupees.

3. Display the Product ID, Price and Quality Index of all the products based on the Defect rate.

4. Display the ID, Name and Price of products whose defect rate is greater than 8.

5. Find the number of products, total quantity, total price value, maximum price, minimum price, and average price.

6. Write a query to retrieve the number of products for each price category and their average defect rate.

Q5. Suppose relations [1]

R(A,B,C) and S(A,B,C) have the following five and four tuples respectively:

A	B	C
	2	3
4	2	3
4	5	6
2	5	3

Compute $R - (S - R)$. [1]

	B	C
2	5	3
2	5	4
4	5	6
1	2	3